UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
International General Certificate of Secondary Education

**BIOLOGY** **0610/13**

Paper 1 Multiple Choice **May/June 2011**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **17** printed pages and **3** blank pages.


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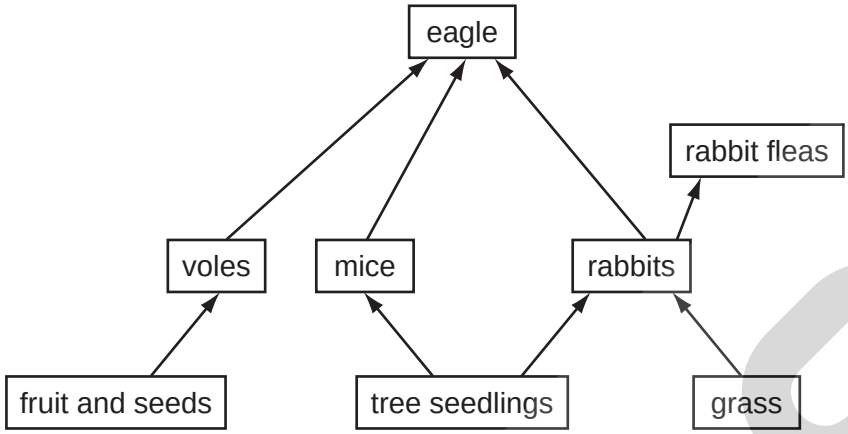
**1** The diagram shows a food web.




Which of the following is a food chain in this web?

**A** grass → rabbits → eagle

**B** grass → voles → eagle

**C** rabbit fleas → rabbits → grass

**D** seeds → voles → mice

**2** In a particular habitat, hyaenas eat antelopes and grass is eaten by antelopes.

What is the source of energy for the grass?

**A** antelopes

**B** carbon dioxide

**C** hyaenas

**D** the Sun

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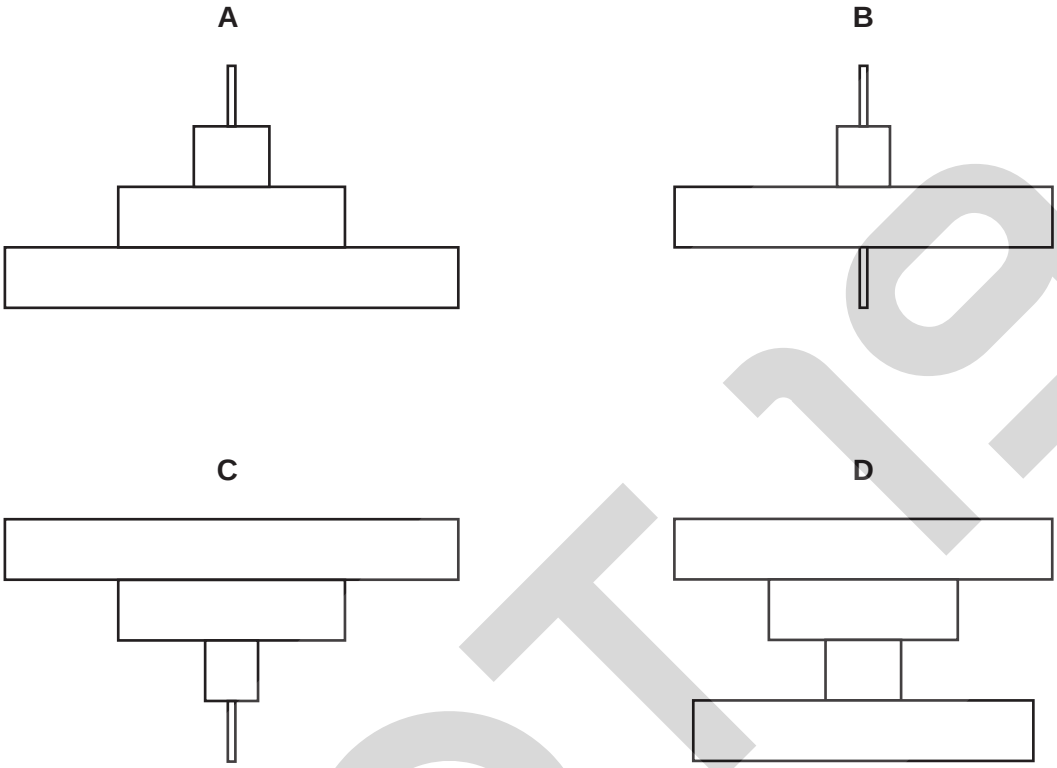
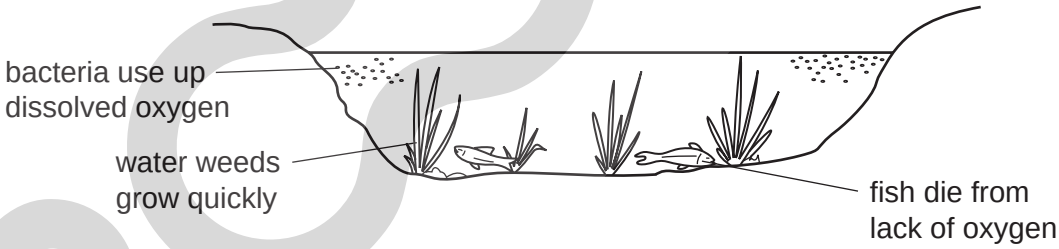
**3** A single tree is food for a large population of caterpillars. Several small birds eat the caterpillars. The small birds are eaten by a bird of prey.

Which diagram shows the pyramid of numbers for this food chain?




**4** The diagram shows some of the effects of human activity on a river.




What could have caused these effects?

**A** air pollution

**B** deforestation

**C** over-use of fertilisers

**D** presence of weed killers

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**5** A large area is heavily overgrazed for many years.

What is the effect on soil nutrients and soil erosion?

| | soil nutrients | soil erosion |
|---|---|---|
| **A** | decrease | decrease |
| **B** | decrease | increase |
| **C** | increase | decrease |
| **D** | increase | increase |

**6** An organism has dry scales covering its body.

To which vertebrate group does it belong?

**A** amphibian

**B** fish

**C** mammal

**D** reptile

**7** The diagram shows an animal.

Use the key to identify the animal.

| | | |
|---|---|---|
| 1 | has legs ..... | go to 2 |
| | has no legs ..... | go to 3 |
| 2 | has a shell ..... | animal **A** |
| | has no shell ..... | animal **B** |
| 3 | has one muscular foot ..... | animal **C** |
| | has more than one foot ..... | animal **D** |

**8** What are enzymes made of?

**A** carbohydrates

**B** DNA

**C** fats

**D** proteins

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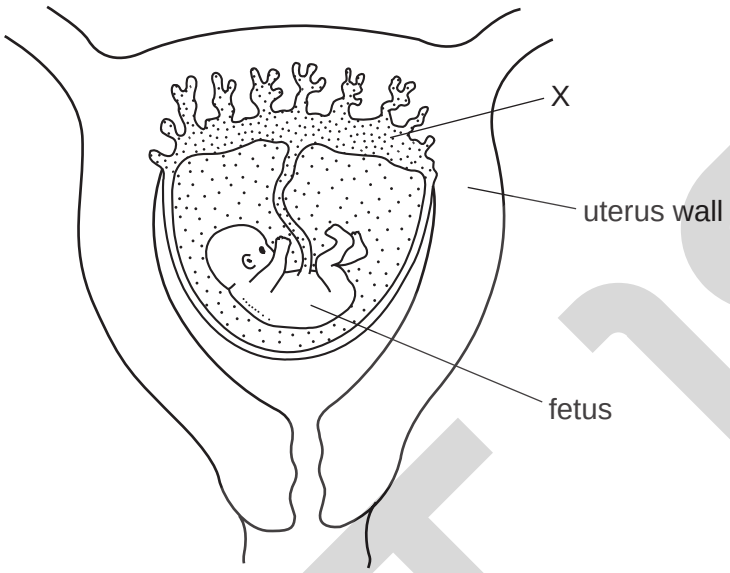
**9** At which temperature do most enzymes from the human body become completely denatured?

**A** 0 °C **B** 27 °C **C** 40 °C **D** 65 °C

**10** The diagram shows a developing human fetus within the uterus.




What is a main function of X?

**A** passing faeces to the mother

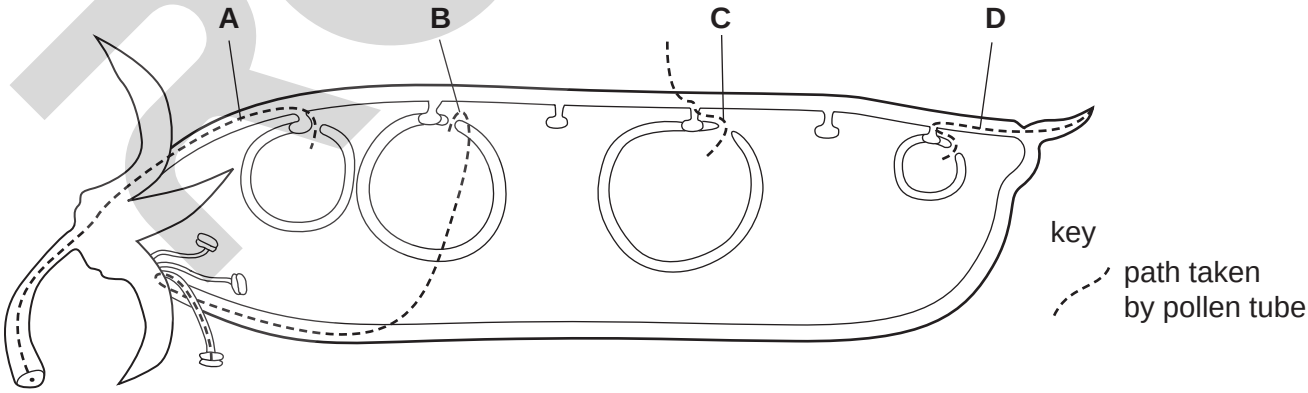
**B** passing oxygen to the fetus

**C** passing the mother's blood to the fetus

**D** protecting the fetus from knocks

**11** The diagram shows a pod from a pea plant.

Which line correctly shows the path that was taken by a pollen tube to an ovule?

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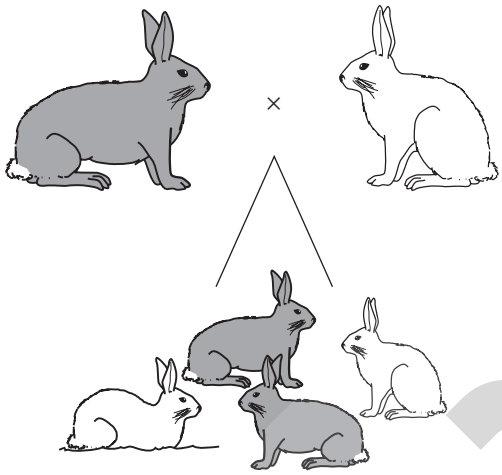
**12** In rabbits, the allele for dark fur, R, is dominant to the allele for white fur, r.

The diagram shows a cross between a rabbit with dark fur and a rabbit with white fur.



What are the genotypes of the offspring?

**A** Rr and rr **B** RR and rr **C** RR and Rr **D** R and r

**13** What is **always** found in female gametes and **may** be found in male gametes?

**A** one X chromosome

**B** one Y chromosome

**C** two X chromosomes

**D** one X chromosome and one Y chromosome

**14** In which order does water pass through these structures in a plant?

**A** mesophyll → root hair → xylem

**B** mesophyll → xylem → root hair

**C** root hair → mesophyll → xylem

**D** root hair → xylem → mesophyll

**15** Which pair of substances is transported in the phloem?

**A** amino acids and protein

**B** amino acids and sucrose

**C** protein and starch

**D** starch and sucrose

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**16** What is formed first in a leaf as a result of photosynthesis?

**A** cellulose

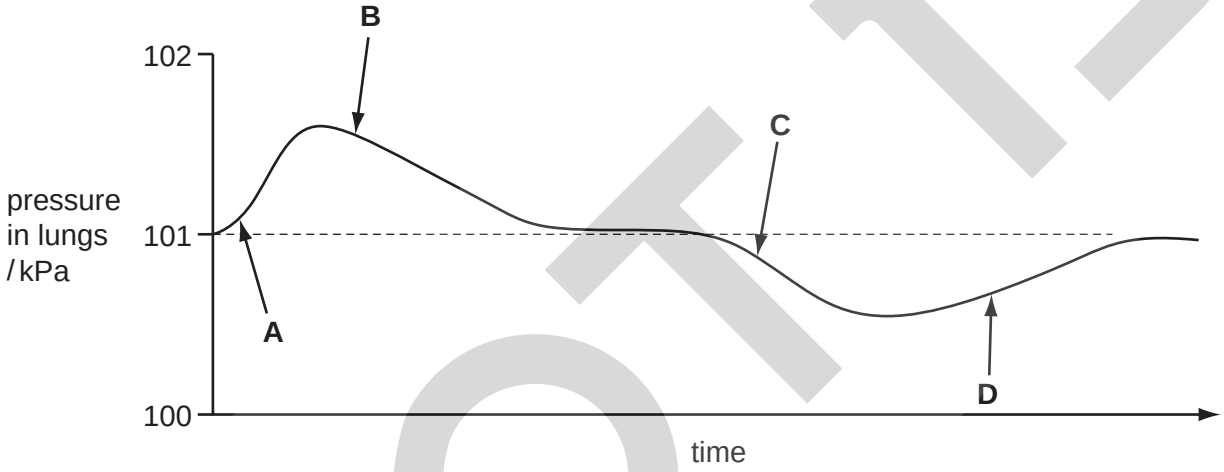
**B** protein

**C** starch

**D** sugar

**17** The diagram illustrates changes in air pressure taking place inside the lungs during a complete cycle of breathing. Atmospheric pressure is 101 kPa.

Which position on the graph corresponds to the point at which the ribs are beginning to be raised?





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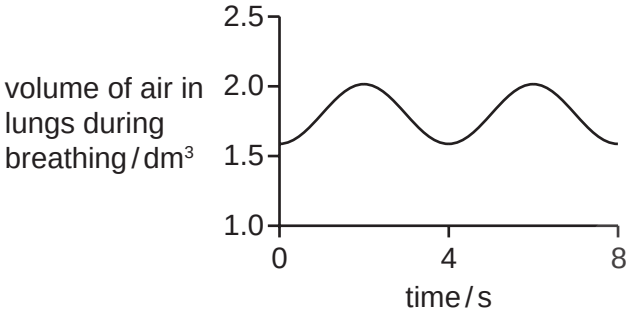
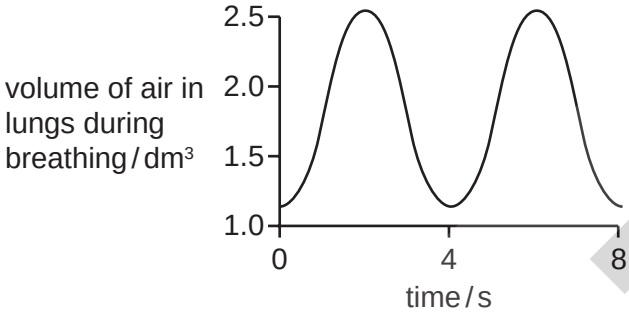
**18** The graph shows the rate and depth of a person's breathing before exercise.

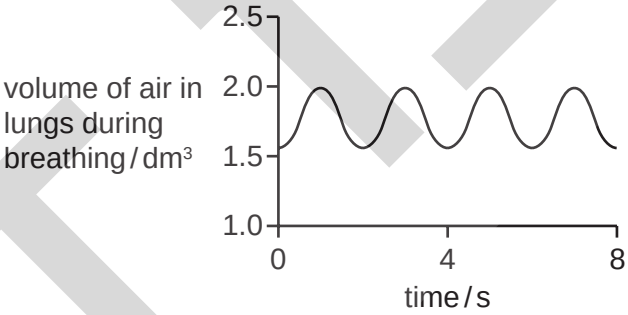



Which graph shows the rate and depth of breathing of the same person immediately after a period of exercise?

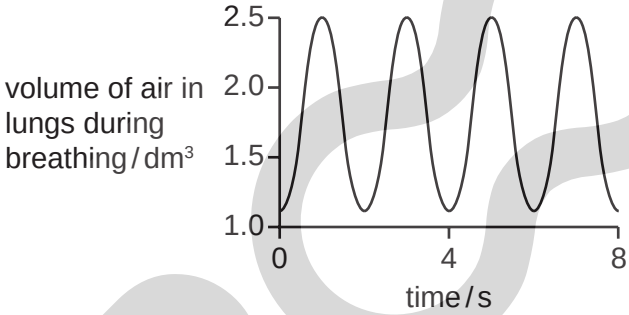
**A**



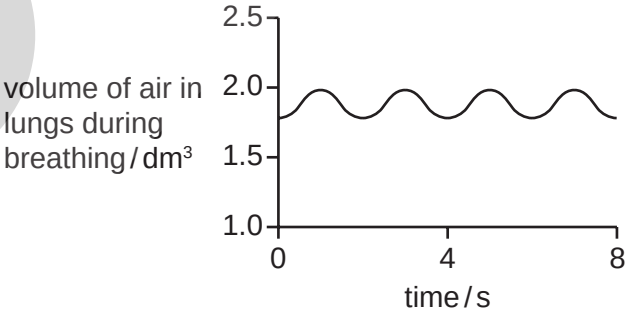

**B**




**C**




**D**

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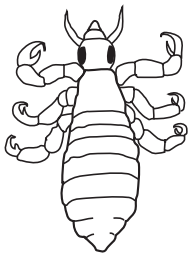
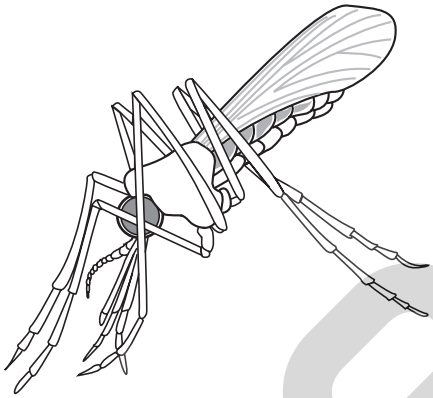
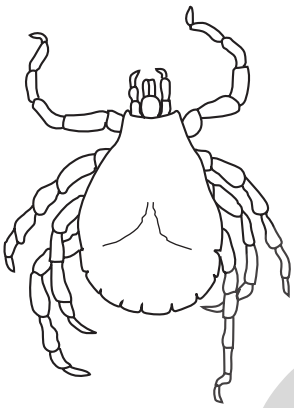
**19** The diagram shows four arthropods.



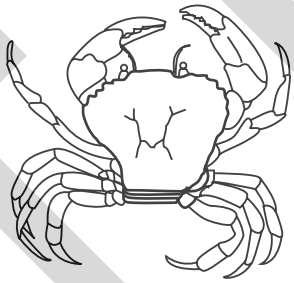
*Pediculus* ×20



*Anopheles* ×10



*Dermacentor* ×7



*Carcinus* ×0.5

How many of these arthropods are insects?

**A** 1 **B** 2 **C** 3 **D** 4

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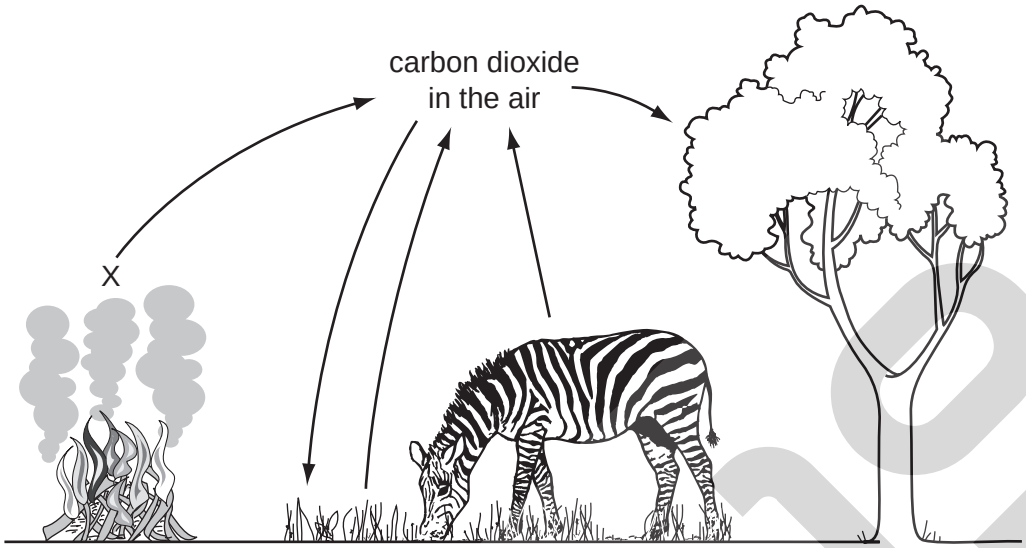
**20** The diagram shows part of the carbon cycle.




Which process is taking place at X?

**A** combustion

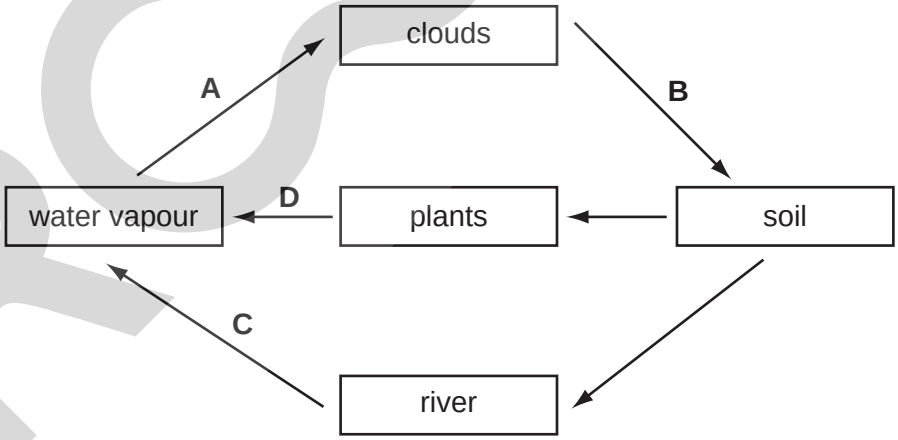
**B** decomposition

**C** photosynthesis

**D** respiration

**21** The diagram shows part of the water cycle.

Which arrow represents condensation?





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**22** Which characteristic is shown when a person detects a gas in the air by its smell?

**A** excretion

**B** movement

**C** respiration

**D** sensitivity

**23** Which statement about respiration is **not** correct?

**A** All living cells respire.

**B** Heat is always produced.

**C** Plants respire in the light and in the dark.

**D** Plants take in carbon dioxide and give out oxygen.

**24** Which process does **not** involve an increase in dry mass?

**A** a bacterium getting larger before it divides

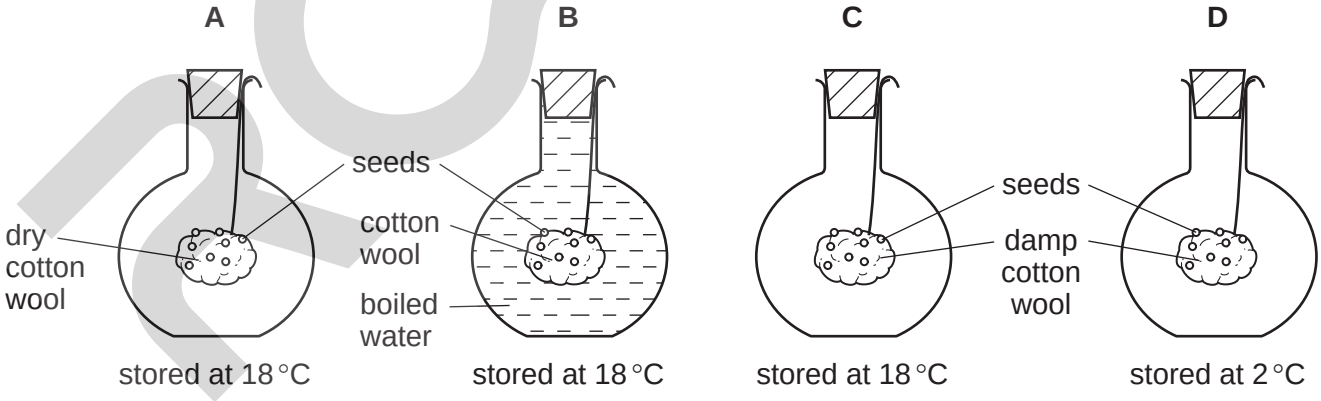
**B** a fetus developing inside the uterus

**C** a green shoot growing towards light

**D** a seed germinating under the ground

**25** The diagram shows four flasks which were set up to investigate the conditions needed for germination.

In which experiment will the seeds germinate most quickly?





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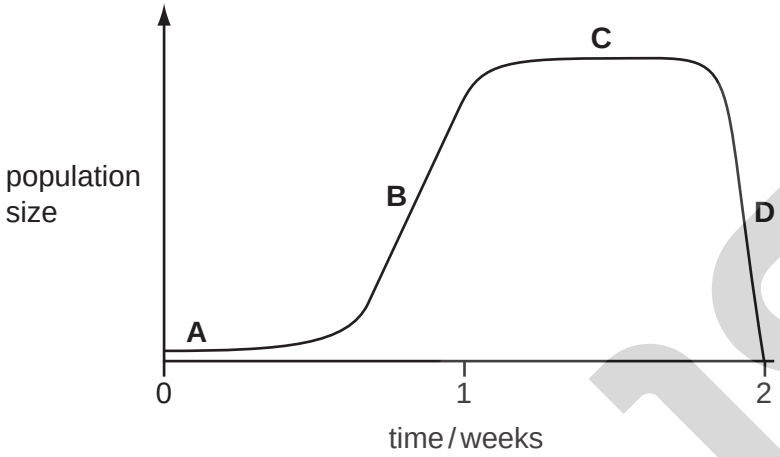
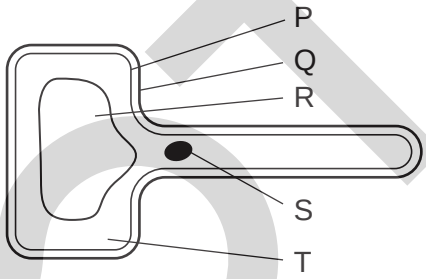
**26** Some bacteria were grown in a nutrient solution over a period of two weeks. The graph shows how the population of bacteria changed during this time.

Which is the lag phase in the growth of this population?




**27** The diagram shows a plant cell.




Which labelled structures are found in plant cells but **not** in animal cells?

**A** P and Q **B** Q and R **C** R and S **D** S and T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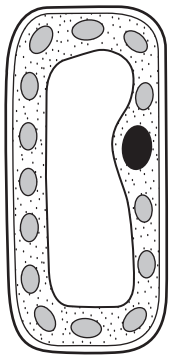
**28** The diagram shows another plant cell.



Which feature indicates that it is a palisade cell?

**A** a large nucleus

**B** a large vacuole

**C** a thick cell wall

**D** many chloroplasts

**29** Which structures are adapted for supporting a plant?

**A** phloem tissues

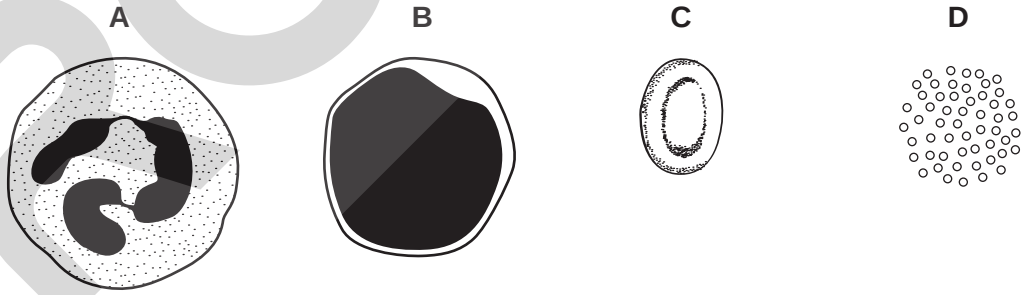
**B** root hair cells

**C** stomata

**D** xylem vessels

**30** The diagram shows some parts from the blood of a mammal.

Which part would contain the breakdown products of bacterial cells?

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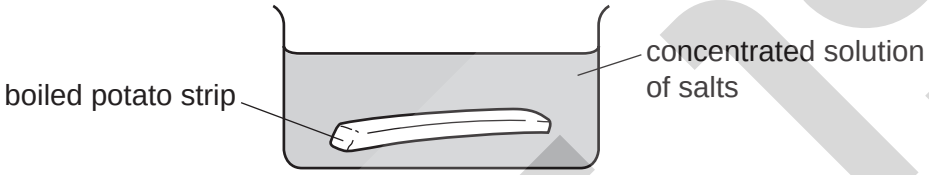
**31** Which describes the structure and function of a red blood cell?

| | structure | function |
|---|---|---|
| **A** | cell contents are dead | transport of water |
| **B** | has a nucleus | produces antibodies |
| **C** | has cilia | moves particles in the respiratory tract |
| **D** | has no nucleus | transport of substances |

**32** Boiling potatoes destroys their cell membranes. A peeled, boiled potato strip is placed in a concentrated solution of salts.




What takes place?

| | osmosis | solute diffusion |
|---|---|---|
| **A** | ✓ | ✓ |
| **B** | ✓ | ✗ |
| **C** | ✗ | ✓ |
| **D** | ✗ | ✗ |

key:
✓ = takes place
✗ = does not take place

**33** A red blood cell is placed in a concentrated sugar solution.

What happens and why?

**A** The cell bursts as sugar molecules diffuse into it.

**B** The cell bursts because the concentrated sugar solution enters it.

**C** The cell shrinks because sugar molecules leave it.

**D** The cell shrinks because water leaves it.

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**34** Nutrients are made up of smaller basic units. Nutrients can be identified by food tests.

Which nutrient is a protein?

| nutrient | smaller basic units | food test |
|---|---|---|
| **A** | amino acids | Benedict's test |
| **B** | amino acids | biuret test |
| **C** | sugars | Benedict's test |
| **D** | sugars | biuret test |

**35** In which part of the body does the breakdown of drugs occur?

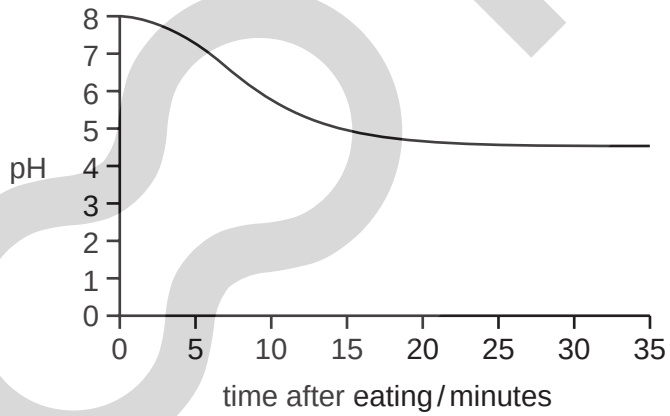
**A** brain

**B** heart

**C** kidneys

**D** liver

**36** The graph shows pH changes in the mouth after eating.




Why is it a good idea to brush teeth after eating?

**A** Acidic conditions help bacteria to grow.

**B** Acids dissolve tooth enamel.

**C** Alkaline conditions help bacteria to grow.

**D** Alkalis dissolve tooth enamel.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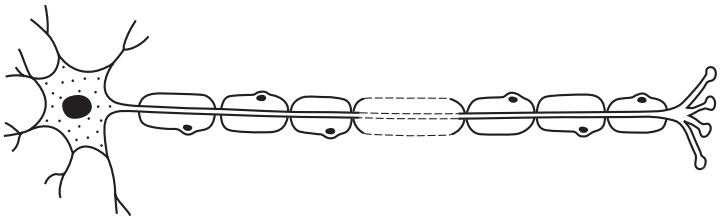
**37** The diagram shows a cell.



What type of cell is shown?

**A** ciliated cell

**B** motor neurone

**C** relay neurone

**D** sensory neurone

**38** The diagram shows four specialised cells.

liver cell  white blood cell  muscle cell  sperm cell

Which feature is **not** common to all of these cells?

**A** cell membrane

**B** cytoplasm

**C** diploid number of chromosomes

**D** nucleus

**39** Weeds are growing in a crop.

What should be used to kill the weeds?

**A** artificial fertiliser

**B** herbicide

**C** magnesium salts

**D** organic manure

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**40** When a bright light is shone into the eye, the diameter of the pupil decreases.

What is this an example of?

**A** accommodation

**B** a simple reflex

**C** photosynthesis

**D** voluntary response

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